ROCK MAGNETISM AND POLAR WANDERING[1]

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ABSTRACT

Lava flows and other igneous rocks become magnetized in the direction of the local geomagnetic field when they cool down after their formation. Similarly, sediments acquire a weak magnetic polarization on deposition. The mean direction of magnetization of series of recent lava flows and sediments has been determined; it is found that these mean directions agree closely with the theoretical dipole field. This field is the field due to a geocentric axial magnetic dipole. The conclusion is therefore drawn that the mean position of the magnetic poles (taken over a period of several thousand years) coincides with the geographic poles.

Assuming that the same is true for the geological past, the position of the geographic poles can be defined within fairly narrow limits by using the mean direction of magnetization of older rocks. Measurements on igneous rocks, of Tertiary and Quaternary age and from Europe only, are available. It is concluded that the large amount of polar wandering suggested by Kreichgauer, Köppen and Wegener, and Milankovitch cannot be reconciled with the new data. If polar wandering has taken place at all, it has not exceeded 5°–10° since Eocene times.

INTRODUCTION

It is known that, on cooling, lava flows become permanently magnetized in the direction of the local geomagnetic field. Chevallier (1925) determined the directions of the permanent magnetization of lava flows on Mount Etna of which the dates were known and was able to trace the changes in direction from the twelfth to the seventeenth centuries; the later results agree with observatory determinations. Minakami (1941) reports that the Yoridai-Sawa lava flow in Japan (erupted on July 12, 1940) actually became magnetized in the direction of the local geomagnetic field. Hospers (1953–1954) came to the same conclusion for the historic lava flows of Mount Hekla, Iceland. Studies of the direction of the permanent magnetization of lava flows of various geological ages have therefore been used to trace the changes in direction of the geomagnetic field in the past. One point which deserves particular attention in this connection is the question of the stability of the natural permanent magnetization of igneous rocks. It is obvious that various processes may alter the direction and intensity of the permanent magnetization of the original (so-called "thermo-remanent") magnetization, acquired on cooling. The rocks may gradually become demagnetized by earth tremors, recrystallization, and weathering. New permanent magnetization may be acquired by reheating (due to deep burial or intrusions in the neighborhood) and subsequent cooling or by lightning. Faulting may also have some influence. It is therefore advisable to investigate the stability of the magnetization. Two field tests, which were devised by Graham (1949), serve this purpose.

The first test is to investigate the directions of the permanent magnetization of a given series of lava flows in a number of sections with different directions and angles of geological dip of the strata. The scatter in the directions of permanent magnetization before and after the correction for the geological dip of the strata is applied is compared. This is done by giving each direction of magnetization (of an individual specimen or, alternatively, of the mean direc-

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tion of magnetization of an individual lava flow) unit weight; the mean direction is consequently the resultant of a number of unit vectors. The lengths of the resultant vectors before and after the correction for the geological dip (the application of this correction may be called "unfolding," in analogy with sedimentary rocks, where samples are taken from the same folded sedimentary bed) are then compared. If "unfolding" increases the length of the resultant vector, the scatter decreases, and this is taken as an indication that the permanent magnetization originated before the flows were dislocated.

The other test is to investigate the directions of the permanent magnetization of individual pebbles in a conglomerate. Their permanent magnetization originated when they were part of a cooling lava flow and has been preserved throughout the subsequent disintegration of the flow and the transport of the fragments. The pebbles were deposited without reference to their directions of magnetization; consequently, when it is found that the directions of magnetization of pebbles in a conglomerate are randomly distributed, it follows that the directions of the permanent magnetization of these pebbles have been stable since deposition. It may therefore be assumed that the directions of permanent magnetization of lava flows of the same petrological character have been stable in direction over comparable periods of time.

Other considerations, such as the decreasing intensity of the permanent magnetization of lava flows with increasing age, the high coercivity, and the great resistance against mechanical maltreatment, may also indicate that the direction of the permanent magnetization has been stable over long periods of time. Most of the above-mentioned criteria have been used by Hospers (1953, 1953–1954) to demonstrate the directional stability of the permanent magnetization of Icelandic lava flows, ranging in age from Miocene (approximately 20×10^6 years ago) to A.D. 1947–1948, the date of the latest eruption of Mount Hekla.

The tests of the stability of the directions of permanent magnetization cannot always be applied or have not always been applied in studies of rock magnetism. Nevertheless, the present author is confident that all plateau basalts and similar fine-grained basic rocks (whether they occur as lava flows, dikes, or sills) which closely resemble the Icelandic lava flows have faithfully preserved their original permanent magnetization over periods of tens of millions of years, provided, of course, that faults, intrusions, and weathering have been avoided during sampling.

Similarly, sediments have been investigated with a view to tracing the changes of the geomagnetic field at the site of deposition. It may be emphasized that for igneous rocks and sediments two entirely different mechanisms lead to the development of permanent magnetic polarization. Igneous rocks (or, rather, the magnetite and other ferromagnetic minerals they contain) acquire a permanent magnetization when the rocks cool down through their Curie point (a theoretical explanation of this phenomenon has been given by Néel, 1949). When a sediment is formed, however, the minute, permanently magnetized ferromagnetic particles are lined up during settling in the local geomagnetic field. The resulting permanent magnetization per cubic centimeter is, on the average, one hundred to one thousand times smaller than that of igneous rocks. The question of the stabil-

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ity of the permanent magnetization is of much greater importance for sediments than it is for igneous rocks. Little is known, however, about the causes of the partial or complete instability (which has been proved in several cases) of the magnetization of sediments. Presumably, it depends on the fact that the intensity of the natural permanent magnetization which originated when the sediment was formed is very small relative to its content of ferromagnetic minerals. The intensity of permanent magnetization of Icelandic sediments, for example, is about one hundred times smaller than that of the plateau basalts, whereas the content of ferromagnetic minerals in both is nearly the same.

Sediments may suffer alterations of the direction and intensity of the permanent magnetization through the same causes described earlier for lava flows and in the same intensity as lava flows (as the magnetite content is similar). A possible new magnetization in sediments, however, is much more likely to swamp the original magnetization, because the latter is one hundred times smaller than in lava flows. Nevertheless, there are sediments the magnetization of which has been stable over long geological periods (Graham, 1949).

If used with caution, studies of rock magnetism may therefore yield valuable evidence concerning the direction of the geomagnetic field in the past. However, many instances have been found of igneous rocks with a permanent magnetization which is approximately in a direction opposite to that of the present geomagnetic field. This phenomenon of reverse magnetization has attracted much attention in recent years (cf. Bruckshaw, 1953), and various explanations have been put forward to explain it. One group considers it as due to anomalous magnetic properties of certain rocks which produce reverse magnetization on cooling in a field which had essentially the same direction as the present field. Other explanations invoke reversals of the local or the main geomagnetic field to account for the observations. In order to account for the magnetization of Icelandic lava flows, the present author has been led to assume repeated reversals through 180° of the main geomagnetic field (in fact, a switching of the earth's magnetic poles); this reversal is thought to take place every 250,000–500,000 years.

Whatever the correct explanation of the reverse magnetization found in igneous rocks, their directions can also be used for tracing the direction (irrespective of sign) of the geomagnetic field in the past, since the mean direction of magnetization of series of reversely magnetized lava flows is exactly 180° different from that of series of normally magnetized flows (Hospers, 1953–1954). The mean direction is therefore determined and the exactly opposite direction taken. This brings the results from reversely magnetized rocks in line with those from rocks with normal magnetization. Reversely magnetized sediments are also known, but they are rare. They have not been used in this paper.

POSITION OF MAGNETIC POLES IN RECENT GEOLOGICAL TIMES

Several studies are available of the natural permanent magnetization of comparatively young igneous and sedimentary rocks (lava flows and varved clays, respectively). In general, a fairly large number of directions of magnetization are available, such as the directions of magnetization of individual specimens taken from different lava flows or the mean directions of magnetization (deter-

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mined from several samples) for a number of different lava flows or the mean direction of magnetization of groups of samples (e.g., groups covering 10 or 1,000 years) from varved clays. Each direction is given unit weight, i.e., it is represented by a vector of unit length. The direction and length of the resultant vector are determined by using direction cosines. The direction of the resultant vector is expressed in declination and inclination, the length is used in the statistical treatment of the results. For this latter procedure Fisher's treatment of dispersion on a sphere (Fisher, 1953) has been used. From the number of unit vectors N, the length of the resultant vector R, and the required probability (P, e.g., 5 per cent), a quantity θ, called the (spherical) radius of the circle of confidence, is calculated from the formula (Fisher, 1953, p. 303):

$$1 - \cos\theta = \frac{N-R}{R}\left\{\left(\frac{1}{P}\right)^{1/N-1} - 1\right\}.$$

Assuming that, for a given set of directions θ, calculated for $P = 5$ per cent, appears to be $a°$, we may use this result as follows: *The probability that the true direction of magnetization deviates more than* a° *in direction from the calculated mean direction of magnetization is 5 per cent.* How extremely useful this piece of information is, will be shown later.

TABLE 1*

LAVA FLOWS OF MOUNT ETNA (FROM CHEVALLIER'S 1925 DATA)

(Geographical Co-ordinates of Mount Etna: 37°7 N.L., 15°0 E.L.)

	Declination	Inclination	Angle with Mean Direction
Mean direction of magnetization....	N. 4°3 E.	+55°8	0°
Dipole field.....	N. 0°	+57°1	1°3

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 2°0$, $N = 11$, $R = 10.80$.

The calculated mean direction of magnetization will also be compared with the direction of the so-called (theoretical) dipole field. This is the field due to a magnetic dipole situated at the center of the earth and directed along its axis or, in other words, the field that would result if the magnetic poles coincided with the geographical ones. The declination of the dipole field is, of course, zero. The inclination I is calculated from the formula

$$\tan I = 2 \tan \lambda ,$$

where λ is the geographical latitude of the point on the earth's surface, where the direction of the dipole field is wanted (cf. Chapman, 1951, p. 20).

Some results of studies on rock magnetism are discussed here, using the foregoing methods:

1. Chevallier (1925) investigated the directions of permanent magnetization of a number of lava flows from Mount Etna, Sicily. Ten flows cover the period from A.D. 1911 to 394 B.C. One flow of unknown age is also used here. In table 1 the mean direction of magnetization of these 11 flows is compared with that of the dipole field, calculated from the geographical latitude of the volcano. The calculated radius of confidence shows that the probability is only 5 per cent that the true direction of magnetization will differ by more than 2°0 from the calculated mean direction of magnetization. As the dipole field differs only by 1°3 from the calculated mean direction, it follows that the mean direction of magnetization is not significantly different from the direction of the dipole field (on the 5 per cent probability level).

2. Hospers (1953–1954) determined the directions of magnetization of 8 post-

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glacial lava flows from southwestern Iceland. These flows cover a period of at least 4,000 years. In table 2 the results of the calculations are shown; it will be seen that the mean direction of magnetization again agrees closely with the direction of the dipole field. Statistically, the two directions are not significantly different (on a 5 per cent probability level).

3. Bancroft (1951) investigated the directions of magnetization of varved clays from Prästmon in Sweden. Forty-six groups of specimens covering a period of 1,000 years (between A.D. 0 and 1,000) were measured. In table 3 the mean direction of magnetization is shown. There is again close agreement between the mean direction of magnetization and the direction of the dipole field. The angular difference is not significant (on the 5 per cent probability level).

4. Johnson, Murphy, and Torreson (1948) investigated the directions of the permanent magnetization of varved clays from New England (United States), covering a period of 6,000 years (between 15,000 and 9,000 years ago). Their data have been arranged in six groups; each group covers a period of 1,000 years, and for each group the declination and inclination of the mean direction were derived from their figures 13 and 14 (*ibid.*). In table 4 the mean direction is shown. It appears that the dipole-field direction lies on the circle of confidence (for 5 per cent probability) around the computed mean direction of magnetization. The two directions are therefore just not significantly different (on a 5 per cent probability level).

Before we draw any conclusions from these data, two remarks must be made. The first concerns the method of calculating the mean direction. In table 3 we have seen that the mean direction (calculated by representing each individual direction by a unit vector) is: declination

TABLE 2*

POSTGLACIAL LAVA FLOWS OF SOUTHWESTERN ICELAND

	Declination	Inclination	Angle with Mean Direction
Mean direction of magnetization	N. 1°.1 E.	+73°.8	0°
Dipole field	N. 0°	+76°.3	2°.5

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 8°.2$.

TABLE 3*

VARVED CLAYS FROM PRÄSTMON, SWEDEN (FROM BANCROFT'S 1951 DATA)

(Geographical Co-ordinates of Locality 63°.1 N.L., 17°.7 E.L.)

	Declination	Inclination	Angle with Mean Direction
Mean direction	N. 2°.6 W.	+73°.4	0°
Dipole field	N. 0°	+75°.8	2°.4

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 3°.3$, $N = 46$, $R = 44.92$.

TABLE 4*

VARVED CLAYS FROM NEW ENGLAND (U.S.A.) (FROM JOHNSON, MURPHY, AND TORRESON'S 1948 DATA)

(Mean Geographical Co-ordinates 42°.8 N.L., 72°.4 W.L.)

	Declination	Inclination	Angle with Mean Direction
Mean direction	N. 4°.5 W.	+51°.4	0°
Dipole field	N. 0°	+61°.6	10°

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 10°$, $N = 6$, $R = 5.90$.

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N. 2°.6 W., inclination +73°.4. If, however, the arithmetic mean of declinations and inclinations is taken, the mean direction is found to be: declination N. 0°.1 E., inclination +70°.9. It goes without saying that the first method is the more correct one, and it has therefore been used in this paper.

The second remark concerns the magnetic stability of the rocks used. It can be demonstrated as follows: The stability of the magnetization of the New England varved clays has been satisfactorily established by the "unfolding" procedure (Graham, 1949, p. 137–143). The practically identical varved clays from Sweden may therefore also be considered as stable, and this all the more so as they are considerably younger than the New England varved clays. Another argument in favor of the magnetic stability of varved clays is that they give a picture of changes in the direction of the geomagnetic field which resembles the secular variation in observatory determinations. The Etna lavas give directions, the later ones of which agree with observatory determinations and therefore must have been stable. The Icelandic postglacial flows are practically identical with the Tertiary and Quaternary plateau basalts, the magnetic stability of which can be demonstrated.

In all four cases described in this section the mean direction of magnetization agrees closely with the direction of the dipole field. This leads to the conclusion that, taken over a period of several thousand years, the magnetic poles have centered on the geographic poles in recent geological times.

POSITION OF GEOGRAPHIC POLES DERIVED FROM ROCK MAGNETISM

Assuming that the foregoing conclusion also holds for the geological past (and there is every reason to think that this simple relation between the magnetization of the earth and its axis of rotation is correct, cf. Runcorn, 1953), measurements of the natural permanent magnetization of older rocks allow us to define the position of the geographical pole in the geological past—this, of course, on condition that we may be sure that we are measuring the permanent magnetization which originated with the rock. The procedure is fairly simple. The individual directions (if necessary, after correction for the geological dip due to later tectonic movements of the strata, which restores the position of the bed as it was when it was formed) are each represented by a unit vector in the measured direction. The mean direction is calculated as well as the radius of the circle of confidence for a given probability (following Fisher, 1953). Throughout this paper $P = 0.05$ or 5 per cent is used. The chances are 19 to 1 that the true direction of magnetization (i.e., the theoretical dipole field direction) lies within a cone, of which the calculated mean direction of magnetization forms the axis and the "radius of the circle of confidence" the semiangle. All directions which lie on this cone can be expressed in declination and inclination; and, by using simple spherical trigonometry as well as the dipole formula (which allows the distance on the globe between the point of observation on the earth's surface and the magnetic pole to be calculated), the position of the magnetic (and therefore also geographic) poles can be calculated which corresponds to each direction on the cone. This gives us a locus on the earth's surface (a somewhat deformed small circle), inside which the geographic pole is situated with a probability of 95 per cent (5 per cent that it lies outside the locus).

This procedure may be illustrated by

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an example. In western Iceland an early Quaternary zone of reversely magnetized lava flows occurs. The mean direction of magnetization, the theoretical dipole-field direction, and the radius of the circle of confidence (for $P = 5$ per cent), as well as the mean geographical co-ordinates of the sampled sections, are shown in table 5. We notice that the angular difference between the mean direction (or, rather, its opposite) and the dipole field is less than θ for $P = 5$ per cent. Consequently, the present geographic north pole lies inside "the locus" (as we shall henceforth call the somewhat deformed small circle on the earth's surface which corresponds to the circle of confidence for $P = 5$ per cent) which we are now about to determine.

On a stereographic projection the (opposite of the) calculated mean direction of magnetization is plotted, with its circle of confidence for $P = 5$ per cent. For four points on this circle the declination and inclination are determined. The inclinations are used to calculate the corresponding co-latitudes (i.e., the distances on the globe from our sections in western Iceland to the corresponding position of the magnetic pole) by using the formula $\tan I = 2 \cot \varphi$, where I is the inclination and φ the co-latitude (cf. Chapman, 1951, p. 20). For each point chosen on the circle of confidence, a spherical triangle on the earth's surface can be drawn, the corners of which are constituted by (1) the mean position of our sections in western Iceland, (2) the present geographic north pole, (3) the required position of the magnetic pole. Two sides (the present geographic co-latitude of western Iceland and the calculated magnetic co-latitude) and the included angle (the declination found for each point chosen on the circle of confidence) are known in these triangles. The third side can therefore be calculated: it is the angular distance from the present geographic pole to the required position of the magnetic pole. As the angular distance from western Iceland to the required position of the magnetic pole is also known, the position of the latter can be determined by a geometric construction. A nearly circular closed curve is drawn through the four points thus found on a stereographic projection of the Northern Hemisphere. This is our locus. The position of the geographic pole in early Quaternary times is now defined within narrow limits. The probability is only 5 per cent that the geographic pole was situated outside the locus.

TABLE 5*

EARLY QUATERNARY LAVA FLOWS WITH NATURAL REVERSE PERMANENT MAGNETIZATION FROM WESTERN ICELAND

(Mean Geographical Co-ordinates 64°.6 N L., 22°.0 W.L.)

	Declination	Inclination	Angle with Dipole Field
Mean direction of magnetization	N. 181°.0 E.	−75°.2	180°–1°.5
Opposite direction	N. 1°.0 E.	+75°.2	1°.5
Dipole field	N. 0°	+76°.7	.....

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 6°.9$, $N = 51$, $R = 45.31$.

A similar procedure has been adopted for all other groups of measurements, to be discussed later. In every case four points were selected; they can often be chosen in such a way as to reduce the required calculations to a minimum. These four points define the locus with sufficient accuracy, particularly as it is always known whether the present geographic north pole lies inside or outside the locus.

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SELECTION OF SUITABLE SERIES OF MEASUREMENTS

The very important and interesting measurements on Tertiary and Mesozoic sediments in the United States carried out by Johnson, Murphy, and Torreson (1948) have not been used, because the magnetic stability of these sediments has not been demonstrated. The measurements on Silurian sediments in the United States (Graham, 1949; Tuve, 1950) have not been used either—even though the magnetic stability of these sediments can be demonstrated—as one isolated set of measurements does not allow reliable conclusions to be reached. As to igneous rocks, here the matter of magnetic instability weighs less heavily. Consequently, a number of investigations of the magnetization of fine-grained basic rocks (flows, dikes, and sills) has been used, when obvious causes of magnetic changes—such as faults and intrusions—were avoided. Moreover, of the groups of igneous rocks used, all groups but the fourth are reversely magnetized. Natural reverse magnetization indicates magnetic stability; if these rocks were magnetically unstable, no reverse magnetization would be found. The magnetic stability of the fourth group has been adequately established.

The period of time covered by each group can be estimated only when lava flows are investigated; when dikes and sills are the only source of information, the question is more difficult. Nevertheless, it may be assumed that the formation of a group of dikes and sills does not take place all at once and that a group of samples presumably covers a period easily exceeding the required minimum of several thousand years. Insufficient time coverage constitutes the reason why a certain number of published measurements on single lava flows have not been used. Other work had to be rejected because the published accounts were not sufficiently detailed.

DISCUSSION OF AVAILABLE INFORMATION

Groups of measurements which are suitable for a determination of the position of the geographic pole in the geological past will now be dealt with in order, from young to old. The data have been evaluated as discussed in the preceding section; each set therefore yields a locus within which the pole was situated at the time of formation of these rocks (on a 95 per cent probability level). The computed loci are shown in figure 1.

Early Quaternary lava flows (plateau basalts) from western Iceland (Hospers, 1953–1954).—These flows are reversely magnetized. The required information is shown in table 5. The present geographic north pole lies inside the locus.

Plio-Pleistocene lava flows from the Plateau Central, France (Roche, 1951).—The information is shown in table 6. The present geographic pole lies inside the locus. These rocks are reversely magnetized; the opposite of the mean direction of magnetization is therefore shown in the table.

Mio-Pliocene lava flows from the same region (Roche, 1951), cf. table 7.—These rocks are also reversely magnetized. The present geographic north pole is situated inside the locus.

Miocene lava flows (plateau basalts) from northern and southwestern Iceland, cf. table 8.—This group includes specimens from both normally and reversely magnetized zones; those from the latter have been taken in the opposite direction. The magnetic stability of these flows can be demonstrated (Hospers, 1953–1954). The present geographic north pole lies inside the locus.

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Tholeiite dikes of northern England of Oligocene or Miocene age (Bruckshaw and Robertson, 1949), cf. table 9.—These dikes are reversely magnetized. The present geographic north pole is situated inside the locus.

Oligocene dikes, necks, and sills from the Plateau Central, France (Roche, 1950, and personal communication), cf. table 10.—These rocks are reversely magnetized. The present geographic north pole lies just outside the locus for $P = 5$ per cent.

Eocene lava flows from Northern Ireland (Hospers and Charlesworth, 1954), cf. table 11.—These flows are reversely magnetized. The present geographic north

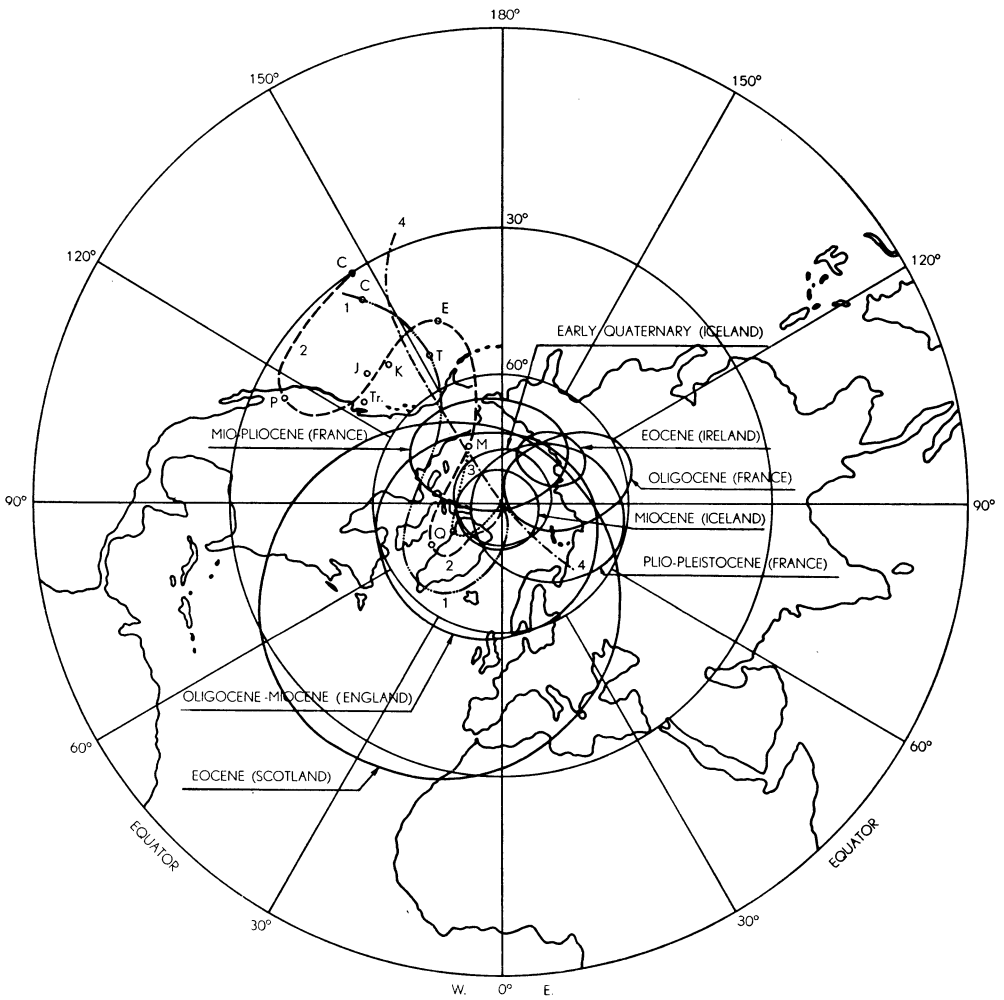


FIG. 1.—Stereographic projection representing the Northern Hemisphere of the earth, with the continents shown in outline. Also are shown the loci calculated from the rock magnetic data, as well as the alleged positions and paths of the geographic north pole in the geological past according to Kreichgauer (curve *1*), Köppen and Wegener (*2*), Köppen (*3*), and Milankovitch (*4*). *C* = Carboniferous; *P* = Permian; *Tr* = Triassic; *J* = Jurassic; *K* = Cretaceous; *T* = Tertiary; *E* = Eocene; *M* = Miocene; *Q* = early Quaternary position of the geographic north pole. These paths were taken from a figure by Köppen (1940), also reproduced by Gutenberg (1951, p. 202, fig. 12).

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pole lies well outside the locus for $P = 5$ per cent.

Eocene lava flows from Scotland (Vincenz, 1952), cf. table 12.—These flows are reversely magnetized. The present geographical north pole lies inside the locus, which is much larger than the one for the contemporaneous flows from Northern Ireland. The reason for this difference is not clear.

TABLE 6*

PLIO-PLEISTOCENE LAVA FLOWS, FRANCE
(DATA FROM ROCHE, 1951)
(Geographical Co-ordinates
45°.1 N.L., 3°.5 E.L.)

	Declination	Inclination
Opposite of mean direction...	N. 17° E.	+62°.6
Dipole field.....	N. 0°	+63°.5

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 12°.8$, $N = 6$, $R = 5.82$.

TABLE 7*

MIO-PLIOCENE LAVA FLOWS, FRANCE
(DATA FROM ROCHE, 1951)
(Geographical Co-ordinates
45°.6 N.L., 3°.0 E.L.)

	Declination	Inclination
Opposite of mean direction..	N. 3°.3 W.	+51°.1
Dipole field.....	N. 0°	+63°.9

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 14°.0$, $N = 5$, $R = 4.85$.

TABLE 8*

MIOCENE LAVA FLOWS FROM NORTHERN AND SOUTHWESTERN ICELAND
(Mean Geographical Co-ordinates
65°.2 N.L., 20°.0 W.L.)

	Declination	Inclination
Mean direction.....	N. 1°.5 E.	+77°.8
Dipole field.....	N. 0°	+77°.0

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 5°.5$, $N = 102$, $R = 88.37$.

HYPOTHESES OF POLAR WANDERING

There is a large variety of polar-wandering hypotheses, and, oddly enough, they all seem to have originated in Europe. Here we shall discuss only some of the better-known ones. One of the oldest forms of this type of hypothesis is that of Kreichgauer (1902). It was proposed on grounds the validity of which is nowadays no longer admitted. A later attempt was that of Köppen and Wegener (1924). They pushed the continents together in the configuration postulated by Wegener's hypothesis of continental drift (Wegener, 1924) and derived the position

TABLE 9*

OLIGOCENE OR MIOCENE THOLEIITE DIKES OF NORTHERN ENGLAND (DATA FROM BRUCKSHAW AND ROBERTSON, 1949, FIG. 7, *B*, P. 316)
(Mean Geographical Co-ordinates
55°.3 N.L., 3°.0 W.L.)

	Declination	Inclination
Opposite of mean direction..	N. 5°.5 W.	+73°.3
Dipole field.....	N. 0°	+70°.9

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 15°.6$, $N = 7$, $R = 6.63$.

TABLE 10*

OLIGOCENE INTRUSIVE ROCKS FROM FRANCE
(DATA FROM ROCHE, 1950, AND PERSONAL COMMUNICATION)
(Geographical Co-ordinates
45°.8 N.L., 3°.1 E.L.)

	Declination	Inclination	Angle with Opposite of Mean Direction
Opposite of mean direction.....	N. 21°.1 E.	+57°.2	0°
Dipole field.....	N. 0°	+64°.1	11°.9

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 10°.8$, $N = 9$, $R = 8.667$.

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of the poles and the equator from the climatic zoning based on paleontological and geological data. This form of the polar-wandering hypothesis has received most attention. Köppen (1940) later suggested some slight modifications in the path of the geographic north pole since Miocene times. It was also Köppen and Wegener's path for the north pole which seemed to be confirmed in a convincing way by Milankovitch's (1933, 1934, 1938) calculations. There is, however, an obvious discrepancy between Köppen and Wegener's assumptions, on the one hand, and those of Milankovitch, on the other: Köppen and Wegener assumed that the continents were much closer together than they are now, whereas Milankovitch assumed that the present configuration prevailed throughout geological times! Also it is now known that if the pole had moved at all along the path calculated by Milankovitch, it is more likely that it would have moved in the opposite direction (Kuiper, 1943; cf. also Umbgrove, 1947, p. 303–304). Nevertheless, Milankovitch's ideas have found some support and have therefore also been included in the present discussion. In figure 1 the paths of the geographic north pole according to these authors are shown. The paths according to Kreichgauer and Milankovitch are in the correct position relative to all continents, as they assumed that the configuration of the continents was the same as the present one for the periods they considered. The positions of the pole according to Köppen and Wegener are, for all periods, in the correct position relative to Africa only; the path of the geographic north pole is in the correct position relative to Europe only for positions of the pole after the Alpine orogeny. The main phase of the Alpine orogeny took place in Middle Oligocene times; in order to get the positions of the pole in pre-Middle Oligocene times in the correct position relative to western Europe, we must undo the shortening of the crust due to the Alpine orogeny. In the present author's opinion this shortening has been systematically overestimated. Nevertheless, we will take a large figure, 600–1,200 km. (5°–10° latitude) (Wegener, 1924, p. 12) for our discussion. This means that Northern Ireland and Scotland, from which we possess Eocene rock magnetic data, should be shifted 5°–10° toward the pole, in order to be in the original alleged position relative to the position of the

TABLE 11*

EOCENE LAVA FLOWS OF NORTHERN IRELAND
(DATA FROM HOSPERS AND CHARLESWORTH, 1954)

(Mean Geographical Co-ordinates of Sections: 55°.1 N.L., 6°.4 W.L.)

	Declination	Inclination	Angle with Opposite of Mean Direction
Opposite of mean direction.....	N. 14°E.	+60°.2	0°
Dipole field.....	N. 0°	+70°.8	11°.5

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 5°.4$, $N = 24$, $R = 23.253$.

TABLE 12*

EOCENE LAVA FLOWS OF SCOTLAND
(DATA FROM VINCENZ, 1952)

(Geographical Co-ordinates 56°.4 N.L., 6°.0 W.L.)

	Declination	Inclination	Angle with Opposite of Mean Direction
Opposite of mean direction.....	N. 43°.1 W.	+78°.1	0°
Dipole field.....	N. 0°	+71°.6	12°.5

* Radius of the circle of confidence: $P = 5$ per cent, $\theta = 23°.7$, $N = 16$, $R = 11.595$.

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geographic pole in Eocene times. We shall return to this matter later.

Milankovitch's calculations did not lead to a correlation of the various calculated positions with the geological time scale. However, he suggested that his calculated path should be considered in the light of Köppen and Wegener's results. We may therefore assume that in Carboniferous times the pole, according to Milankovitch, was in latitude 30°, in Eocene times in latitude 46°, and in Miocene times in latitude 74° (relative to present-day co-ordinates).

A polar-wandering hypothesis of more recent date is the polar-shift hypothesis of Vening Meinesz (1947). As this shift is thought to have taken place in at least pre-Carboniferous and probably in pre-Cambrian times, we shall not discuss it here, as we are concerned only with polar wandering in Tertiary and Quaternary times.

ROCK MAGNETISM AND POLAR WANDERING

It will have been noted that all localities from which rock magnetic data have been used for the calculation of loci are situated in western Europe and Iceland. Both the rock magnetic data and the alleged positions of the poles therefore refer to western Europe and may be compared (cf. fig. 1).

We shall first compare the rock magnetic data with the alleged positions of the north pole, without taking into account: (1) the effect of the Alpine orogeny, (2) possible uncertainties in the age determinations of the rocks, and (3) possible differences in behavior between Iceland and western Europe.

The position of the pole in Tertiary times suggested by Kreichgauer is outside all loci for Tertiary data and must be rejected; the later part of this curve is also well outside the loci for the early Quaternary data from Iceland and the Plio-Pleistocene data from France. The path suggested by Köppen and Wegener must also be rejected for the following reasons. The suggested position of the Eocene pole is far outside the two Eocene loci, and the same is true for the position of the Oligocene pole found by interpolation between the Eocene and Miocene positions. The position of the Miocene pole is inside the locus for the Mio-Pliocene data from France and just inside the locus for the Oligocene-Miocene data of northern England, but outside the Miocene locus of Iceland. The older part of Köppen and Wegener's curve is therefore wrong. The position of the pole in early Quaternary times, according to Köppen and Wegener, and the later improvement by Köppen in the position of the pole in early Quaternary times must also be rejected: they are outside the loci for the early Quaternary data from Iceland and the Plio-Pleistocene data from France. The pre-Miocene part of Milankovitch's curve is also unsatisfactory for similar reasons. The proposed paths must consequently be rejected without exception; the amount of polar wandering suggested by these paths is far too large to be reconciled with these new data.

We may now try to allow for the alleged amount of shortening of the earth's crust during the Alpine orogeny. This means that we have to push Ireland and Scotland to the north over 5°–10° of latitude. With it, we shift the loci over the same distance. It will be seen from figure 1 that this does not bring the alleged position of the Eocene pole inside the loci for the Eocene data, and consequently the objections enumerated against the depicted paths of the pole still hold.

We may now discuss the uncertainties in the age determinations of the rocks

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used. The Miocene lava flows of Iceland might conceivably turn out to be of Oligocene or even Eocene age; the Eocene flows of Scotland and Northern Ireland may be Oligocene or even early Miocene (cf. discussion in Hospers and Charlesworth, 1954). The first change in age would make the incompatibility of the alleged pole paths and the rock magnetic data only more apparent. The second change, if made from Eocene to Oligocene, would still demonstrate the alleged fact, happen. We notice that the present geographical north pole is situated inside all loci except those for the Oligocene data from France and the Eocene data from Northern Ireland. It may be emphasized, first of all, that the loci in figure 1 are based not on certainty but on probability. The loci have all been calculated for a probability level of 5 per cent. This means that, on the average, in 1 out of 20 cases the pole may lie outside the locus. However, in table 13 it is shown

TABLE 13

RADII OF THE CIRCLE OF CONFIDENCE FOR VARIOUS PROBABILITIES, FOR THE OLIGOCENE DATA FROM FRANCE (CF. TABLE 10) AND THE EOCENE DATA FROM NORTHERN IRELAND (CF. TABLE 11)

	Angle between Mean Direction of Magnetization and Direction of Dipole Field	Probability, P, Equals (Per Cent)	Radius of Circle of Confidence, θ, Equals	N	R
Oligocene data from France	11°.9	5	10°.8	9	8.667
	.....	1	14°.0	.....	.....
Eocene data from Northern Ireland	11°.5	5	5°.4	24	23.253
	.....	1	6°.8	.....	.....
	.....	0.1	8°.6	.....	.....
	.....	0.01	10°.2	.....	.....

Oligocene position to be incorrect, with or without allowances being made for the Alpine orogeny for the data from Ireland and Scotland or even for those from northern England and France. If the age is changed from Eocene to Miocene, the Alpine orogeny need not be allowed for, and objections to the alleged Miocene position still remain. As to possible differences in behavior between the European continent and Iceland (which might be considered as a piece of ocean floor above sea-level) very little can be said.

We may use the rock magnetic data not only to establish what has not happened but also to find out what did, in that even for $P = 0.01$ per cent the mean direction of magnetization of the Eocene basalts of Northern Ireland is still outside the circle of confidence (the mean direction for the Oligocene rocks from France is already inside the circle of confidence for $P = 1$ per cent). There is therefore some reason to believe that the position of the geographic pole in Eocene time relative to Ireland was somewhat different from its present position.

One may wonder whether there is any difference between the results from Continental localities and those from Iceland. The available information does not suggest any such difference, but the question could be decided only if much more in-

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formation were available and particularly if Eocene data from Iceland could be used. If any such difference could be found, it might indicate that the apparent small shift of the geographic pole deduced from the Eocene data from Ireland is not due to a wandering of the poles but to a drifting of the European continent relative to the ocean floor.

Finally, it may be mentioned that, if loci for $P = 5$ per cent were constructed for the data used earlier in this paper (lava flows of Mount Etna, postglacial flows of southwestern Iceland, Pleistocene varved clays from Sweden and the United States), the present geographic pole would lie inside (in the last-mentioned case, on) the locus.

In conclusion it may therefore be said that the amount of polar wandering in Tertiary and Quaternary times suggested by the authors of various polar-wandering hypotheses is far too large to be reconciled with our data. However, this does not necessarily mean that no polar wandering or continental drift has taken place at all. There may have been a small shift of the pole or a small drift of the European continent relative to the ocean floor and the pole (the two processes are at present indistinguishable) amounting to 5° or 10°. Only more data can decide this matter. The conclusion that no large amount of polar wandering took place in Tertiary and Quaternary times supports the similar conclusions reached by Berry (1929) from a study of Arctic fossil floras; by Chaney (1940) from a study of plant-bearing deposits of Eocene age; by Behrmann (1939, 1944) from geomorphological evidence; by Schwarzbach (1946, 1950) from paleoclimatological data; by Durham (1950) from a study of marine climates of the Pacific Coast of the United States; and by Umbgrove (1947, p. 276–281, 302–308) from a critical general discussion of the problems.

ROCK MAGNETISM AND CONTINENTAL DRIFT

The idea that rock magnetic studies might eventually help to solve the problem of continental drift was first expressed by Mercanton (1926) and in later years by Gutenberg (1940, p. 448; 1951, p. 204) and Graham (1949, p. 160). To this end we shall have to possess much more information than we do at present. We need rock magnetic data from as many continents as possible and from as many different ages as possible, going back as far as the Carboniferous. It may be expected that igneous rocks will be more suitable than sedimentary rocks for this purpose; but even for igneous rocks it seems doubtful whether they will have been able to preserve their original magnetization over such enormous periods of time. Another difficulty inherent in the rock magnetic approach is that changes in geographical longitude cannot be detected.

King (1953) has recently announced the possibility of a new test of the hypothesis of continental drift, consisting of a stratigraphic comparison of South Africa and Queen Maud's Land (Antarctica). The present author would suggest that a rock magnetic comparison of the Drakensberg plateau basalts and the plateau basalts of Queen Maud's Land (which are thought to be stratigraphically equivalent) might give valuable information. The difference in latitude between the two series before drift occurred cannot have been more than a few degrees, whereas at present it amounts to approximately 30°. However, it does not seem certain that the method can be applied, as the age of these plateau basalts is considerable.

CONCLUSIONS

It is concluded that if any polar wandering has taken place at all in Tertiary

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and Quaternary times, the shift of the poles is far smaller than that suggested by Kreichgauer, Köppen and Wegener, or Milankovitch.

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